

Radioisotope Tracers

- Background info
 - Radioactive decay - concepts and equations
 - Radiometric dating
 - Secular equilibrium
- Case studies
 - ^{14}C
 - Water Mass Tracer
 - $^{238}\text{U} \rightarrow ^{234}\text{Th}$
 - Mixed-layer export via sinking particles

Radioisotopes and decay

Definitions

Parent – Original Radioactive Atom

Daughter – The Product of Decay

Decay Chain – A Series of Decays

Common modes of radioactive decay

Type	Emitted Particle	Δ protons	Δ neutrons	Δ atomic mass	Parent $\rightarrow$ daughter movement on periodic table	Example:
Alpha α	He^{2+}	-2	-2	-4	2 positions to the left	$^{238}\text{U} \rightarrow ^{234}\text{Th}$
Beta β^-	e^-	+1	-1	0	1 position to the right	$^{14}\text{C} \rightarrow ^{14}\text{N}$
β^+ & e^- capture (rare)	e^+	-1	+1	0	1 position to the left	$^{23}\text{Mg} \rightarrow ^{23}\text{Na}$
K-capture (rare)	None	-1	+1	0	1 position to the left	$^{40}\text{K} \rightarrow ^{40}\text{Ar}$

Decay is a property of the nucleus, and independent of chemistry, temperature, and pressure.

Radioactive Decay

Equation describing decay of a parent isotope to a daughter

$$\frac{dN}{dt} = -\lambda N$$

where:

N = the number of atoms of the radioactive substance present

λ = the first order decay constant (t^{-1})

Integration from $t=0$ to later time t gives:

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t \quad \text{or} \quad N = N_0 e^{-\lambda t}$$

where:

N_0 = number of atoms initially present

N = number of atoms present at time t

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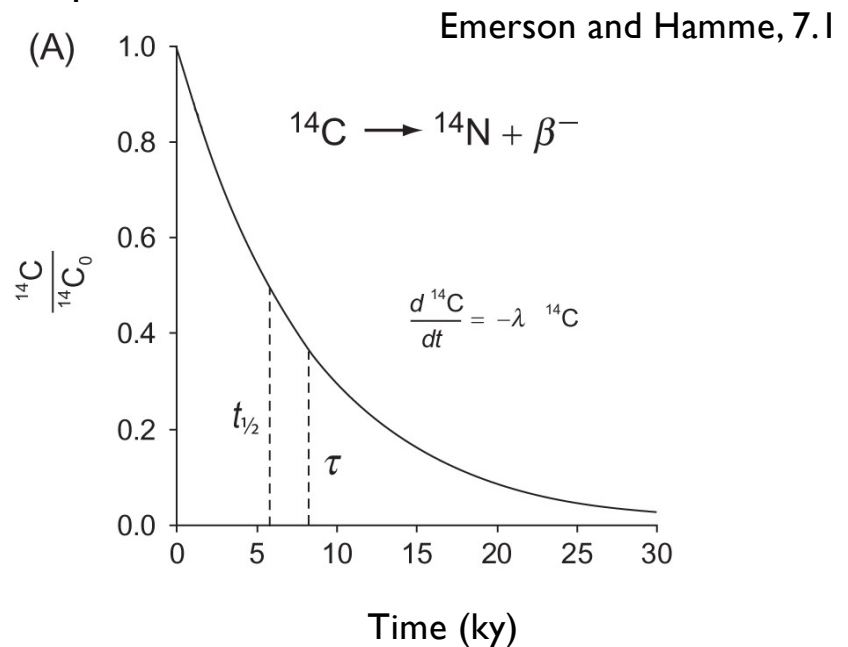
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N = number of atoms present at time t



Radioactive Decay

Activity (A) = decays per time (dpm or dps) –

$$A = -\frac{dN}{dt} = \lambda N$$

λ = decay constant (t^{-1})

N = # of atoms (atoms) or concentration (atoms L^{-1})

Decay is proportional to the # of atoms present (first order)

Units:

Becquerel (Bq) = 1 dps (disintegrations per second)

Curie (Ci) = 3.7×10^{10} Bq = Activity of 1 gram of ^{226}Ra

- Activity is easier to measure than total number of atoms that have decayed

Radioactive Decay

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$$A = -\frac{dN}{dt} = \lambda N$$

λ = decay constant (t^{-1})

N = # of atoms (atoms) or concentration (atoms L^{-1})

From: $N = N_0 e^{-\lambda t}$

we can multiply both sides by the decay constant to get:

$$A = A_0 e^{-\lambda t}$$

- Activity is easier to measure than total number of atoms that have decayed

Half Life

The half life is defined as the time required for half of the atoms initially present to decay.

After one half life: $\frac{N}{N_0} = \frac{1}{2}$

Since: $N = N_0 e^{-\lambda t}$

$$-\ln\left(\frac{1}{2}\right) = \lambda t_{1/2} \quad \text{or} \quad 0.693 = \lambda t_{1/2}$$

$$t_{1/2} = \frac{0.693}{\lambda}$$

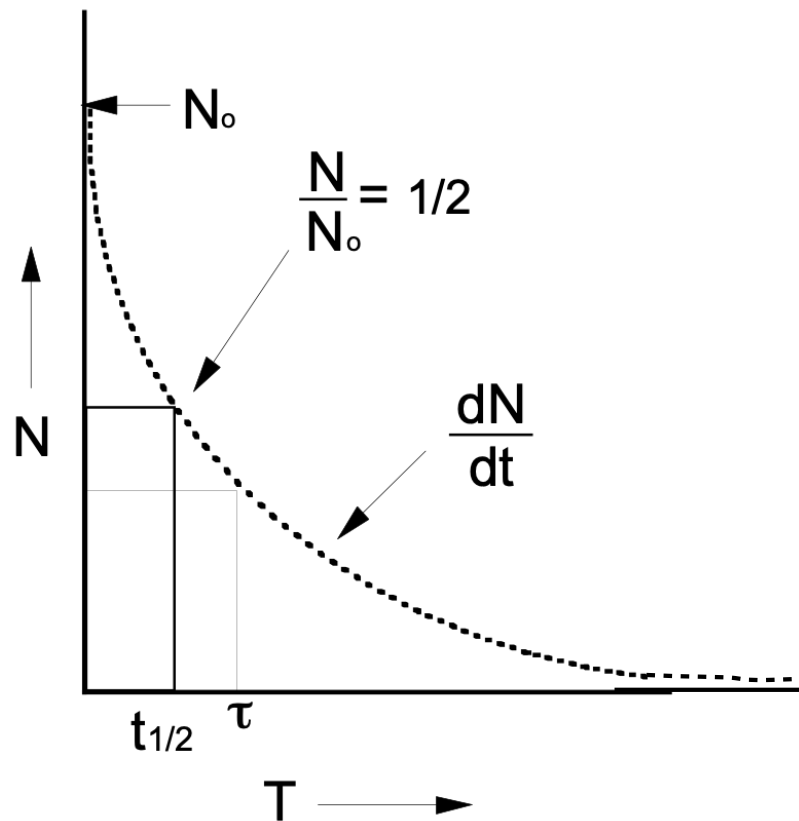
Mean Life, τ

i.e., Average Life of an Atom prior to decay

Mathematically, it is the integral of all lives of the atoms in a particular nuclide divided by the initial N (i.e., N_0):

$$\tau = \frac{1}{N_0} \int_0^{\infty} t dN = \frac{1}{N_0} \int_0^{\infty} t \lambda N dt = \lambda \int_0^{\infty} t e^{-\lambda t} dt = \left[\frac{\lambda t + 1}{\lambda} e^{-\lambda t} \right]_0^{\infty} = \frac{1}{\lambda} = \frac{t_{1/2}}{0.693}$$

Decay curve



Mean Life (i.e.,
Average Life of an
Atom):

$$\tau = \frac{1}{\lambda} = 1.44 \cdot t_{1/2}$$

Q. Why is the mean
life longer than the
half life?

Both N and A decrease exponentially

Radiometric Dating

If there are no other sources and sinks of N ,

$$N = N_0 e^{-\lambda t}$$

If the daughter product of the radioactive decay is stable, the amount of daughter product (N_D) produced by decay is:

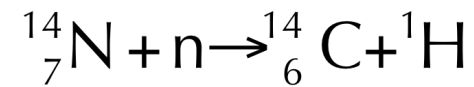
$$N_D = N_{P0} - N_P$$

$$\text{Since } N_P = N_{P0} e^{-\lambda t}, \quad N_D = N_{P0} (1 - e^{-\lambda t})$$

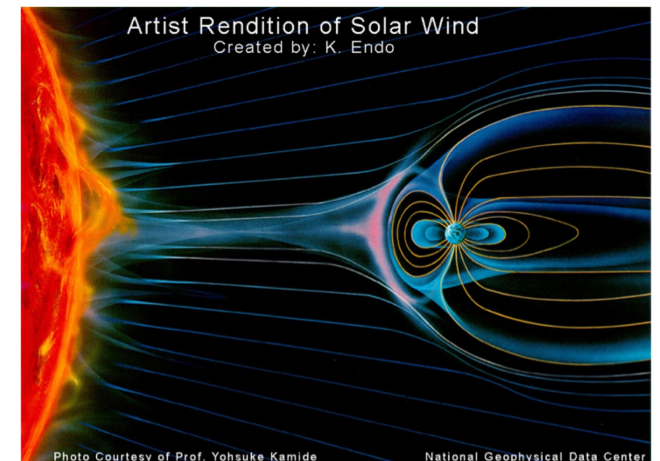
This all assumes that the daughter product is stable.

^{14}C - A quick primer

- ^{14}C formed primarily in the stratosphere via neutron capture by ^{14}N :



- Production of ^{14}C is influenced by:
 - Flux of cosmic rays (decadal-centennial scale)
 - Geomagnetic shielding (centennial-millennial scale)
- Rapidly oxidized to $^{14}\text{CO}_2$



Useful Equations

$\delta^{14}\text{C}$, $\Delta^{14}\text{C}$

Definitions:

$$\delta^{14}\text{C} = \left(\frac{{}^{14}\text{C}/\text{C}_{\text{Sample}}}{{}^{14}\text{C}/\text{C}_{\text{Standard}}} - 1 \right) \times 1000$$

$$\Delta^{14}\text{C} = \delta^{14}\text{C} - 2(\delta^{13}\text{C} + 25) \left(1 + \frac{\delta^{14}\text{C}}{1000} \right)$$

where:

$$\delta^{13}\text{C} = \left(\frac{{}^{13}\text{C}/{}^{12}\text{C}_{\text{Sample}}}{{}^{13}\text{C}/{}^{12}\text{C}_{\text{Standard}}} - 1 \right) \times 1000$$

Characteristics:

$$\begin{aligned} {}^{14}\text{C}/\text{C} \text{ (for } \Delta^{14}\text{C} = 0) \\ = 1.18 \times 10^{-12} \end{aligned}$$

$$t_{1/2} = 5730 \text{ y}$$

$$\tau = 8270 \text{ y}$$

$$\lambda = 1/8270 \text{ y}^{-1}$$

A “conventional ^{14}C age” is defined as:

$$t = -8033 \ln (\text{pMC}), \text{ where } \text{pMC} = (A_{\text{SN}}(1950)/A_{\text{ON}}(1950))$$

- pMC = Percent Modern Carbon
- 8033 years is the mean ^{14}C lifetime based on the half-life of 5568 years
- A_{SN} and A_{ON} are the fractionation-corrected specific ^{14}C activities of the sample and standard, respectively, measured in AD 1950

pMC of 1.0 is equivalent to:

- $1.176 \times 10^{-12} \text{ }^{14}\text{C}/\text{C-atoms}$ (at $-25\text{‰ } \delta^{13}\text{C}$)
- 13.56 dpm/g-C
- 226.0 $\mu\text{Bq}/\text{mg-C}$

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Notes:

- pMC mixes linearly, while ^{14}C age will not!
- $t = 0$ in 1950
- pMC uses the “Libby” decay constant of $1/8033 \text{ yr}^{-1}$, which is different than “Cambridge” decay constant given earlier ($1/8270 \text{ yr}^{-1}$)

Conventional ^{14}C age vs. Calendar Age

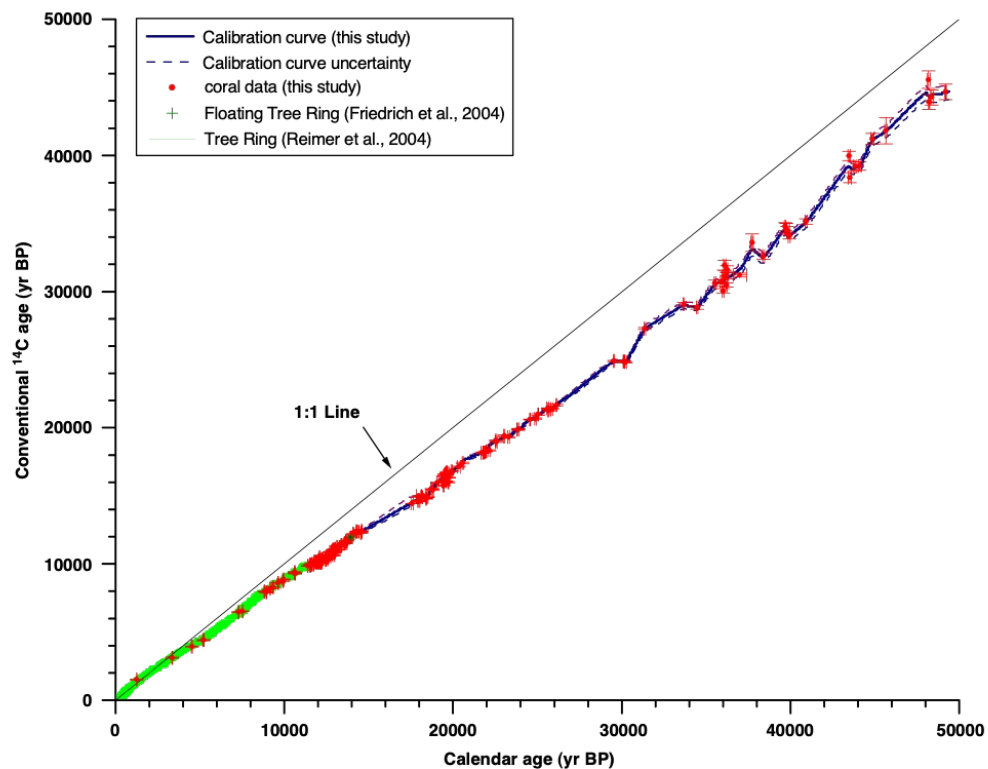


Fig. 3. Our coral calibration curve plotted with 1% confidence limits (corresponds to 3σ uncertainties for normal distributions) and coral data plotted with 1σ error bars. Calibration curve is compared to the tree ring chronologies (Reimer et al., 2004; Friedrich et al., 2004).

Conventional ^{14}C age vs. Calendar Age

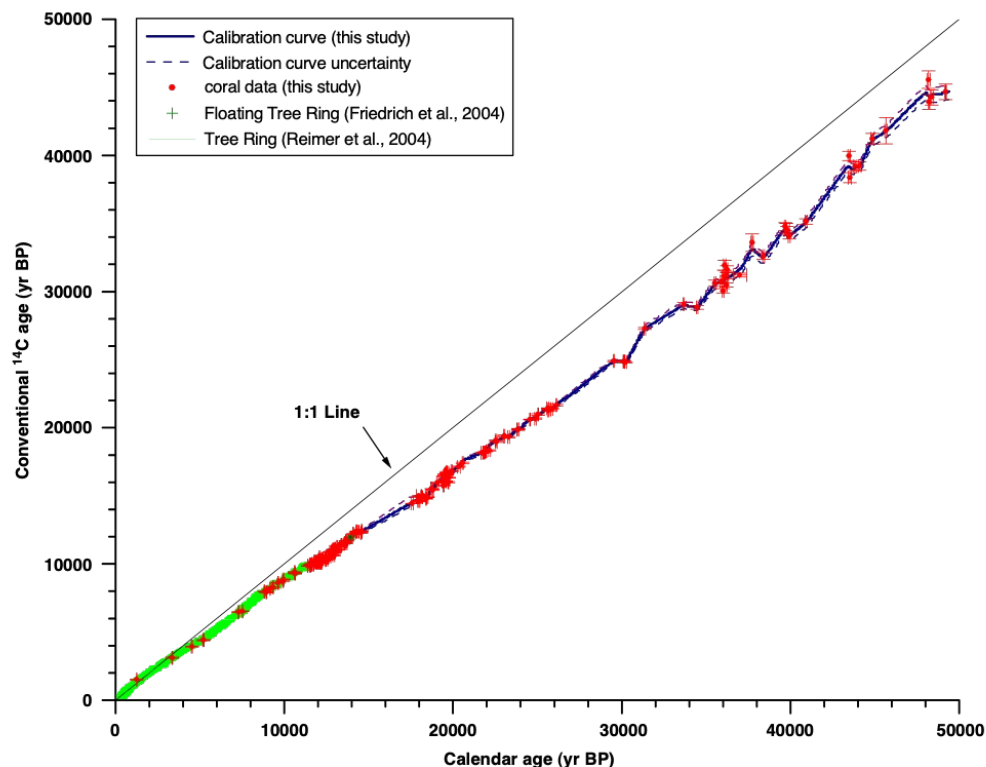


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Fairbanks et al., 2005

Reservoir effect

Radiocarbon samples obtain their carbon from a different source (or reservoir) than atmospheric carbon.

The average difference between a radiocarbon date of a terrestrial sample (e.g., wood), and marine CaCO_3 is about **400 radiocarbon years**. This apparent age of oceanic water is caused both by:

- The delay in exchange rates between atmospheric CO_2 and ocean bicarbonate.

- Dilution effect caused by the mixing of surface waters with upwelled deep waters which are very old.

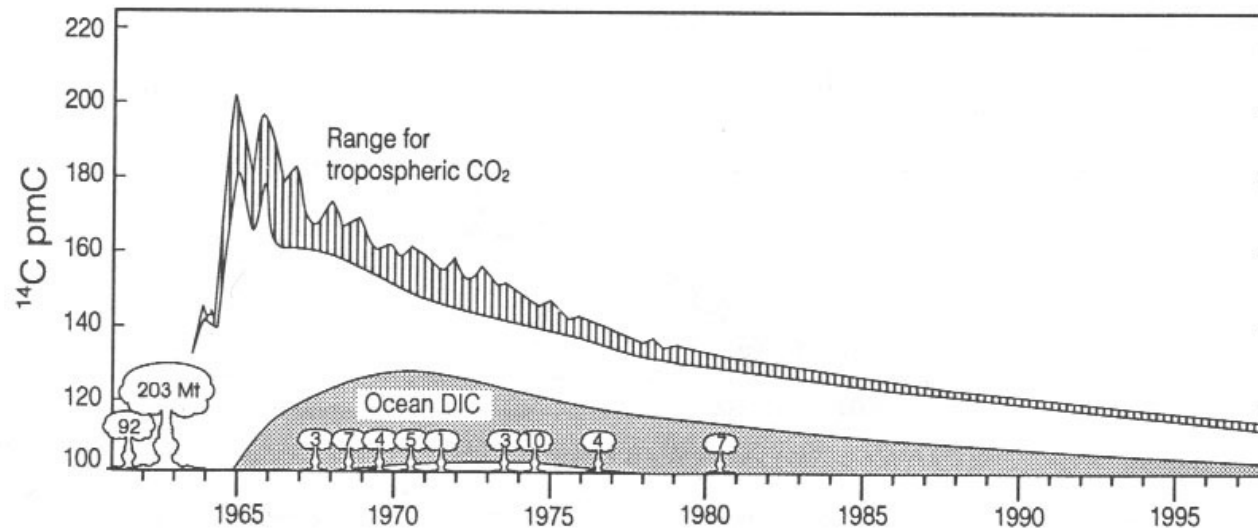
Suess effect

Since about 1890, emissions of radiocarbon-free fossil fuel CO_2 into the atmosphere has **diluted** the the atmospheric radiocarbon signal by about **2%**.

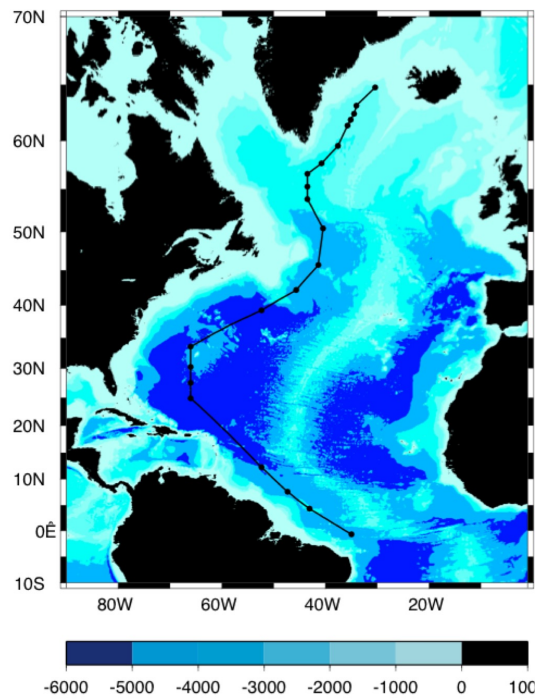
Atom bomb effect

Thermonuclear tests produced 'bomb' ^{14}C , which almost double the amount of ^{14}C activity in terrestrial carbon bearing materials. In the northern hemisphere the amount of artificial carbon in the atmosphere reached a peak in 1963 (in the southern hemisphere around 1965) at about 100% above normal levels.

^{14}C in tropospheric CO_2 and in DIC of the surface ocean



- ^{14}C content of the northern hemisphere increased almost twofold during the mid-1960s as a consequence of the atmospheric testing of thermonuclear bombs (annual total testing given in megatons).
- Increase in southern hemisphere was lower
- Since about 1970 the concentrations have been very similar around the globe.
- Oceanic ^{14}C increased through the uptake of atmospheric ^{14}C .

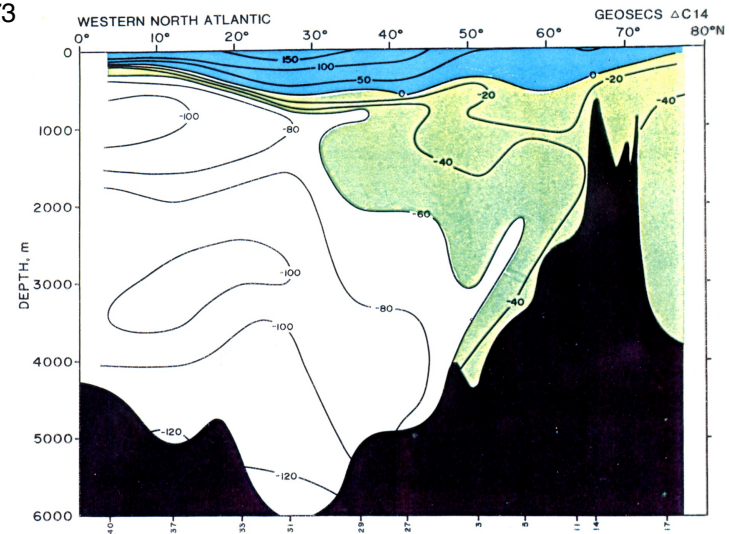


Bomb $^{14}\text{C} = \Delta^{14}\text{C} > -50\%$

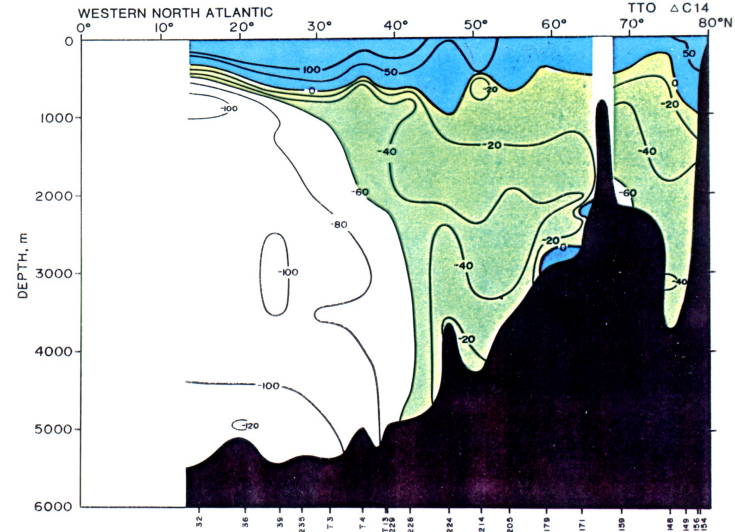
Rule of thumb for comparing $\Delta^{14}\text{C}$ and age:

- 10% decrease in $\Delta^{14}\text{C}$ is roughly 80 year in increase in age

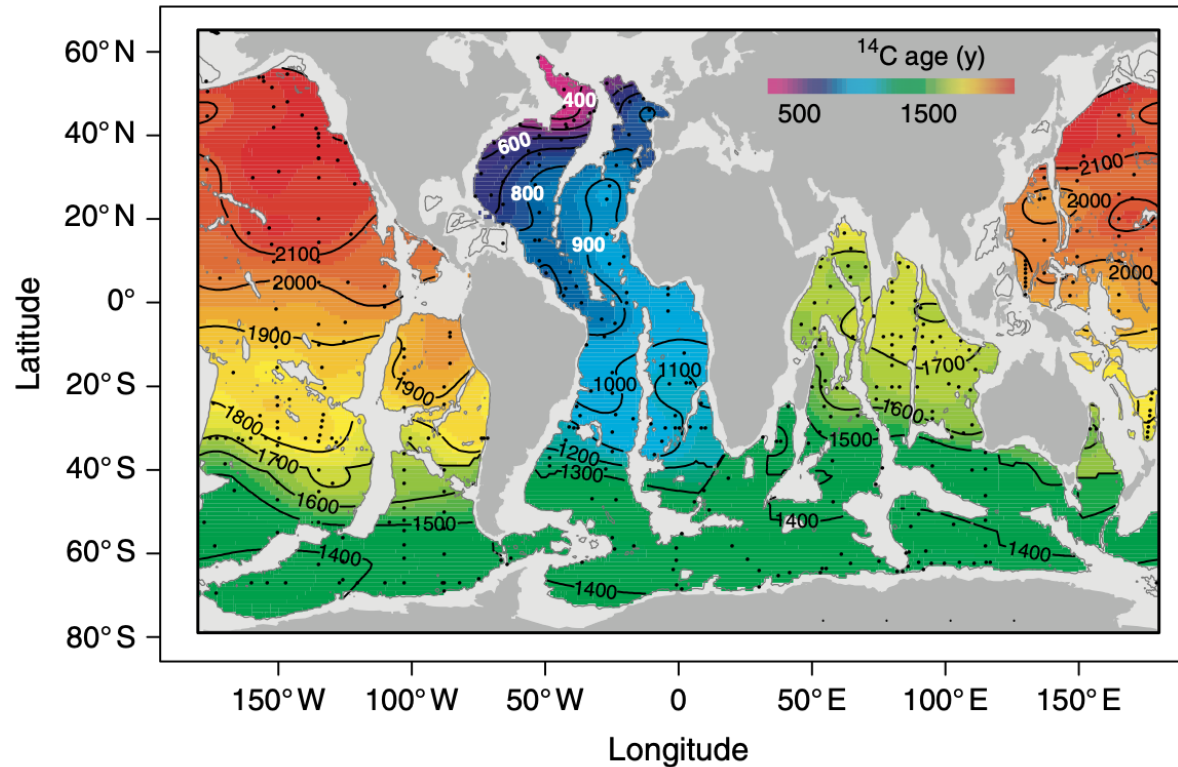
1972-1973



1981



^{14}C age of DIC at 3000 m depth, determined during WOCE



Note:

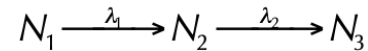
- ^{14}C age of surface ocean is 400 years, because of incomplete air-sea exchange with atmosphere, and due to supply of old, deep carbon from below.

Secular Equilibrium

- The activity of a radioactive parent and daughter will eventually become equal
 - Assumes closed system
 - Assumes parent and daughter stay in contact
- Departures from this equilibrium can be used to calculate fluxes or rates

Secular Equilibrium

Consider a system of three isotopes:



N_1 : Parent

N_2 : First daughter

N_3 : Second daughter

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 \left(e^{-\lambda_1 t} - e^{-\lambda_2 t} \right) + N_2^0 e^{-\lambda_2 t}$$

For $\lambda_2 \gg \lambda_1$, ($t_{1/2,2} \ll t_{1/2,1}$) the $e^{-\lambda_2 t}$ terms decay very fast.

After about 5 mean-lives of N_2 , we have:

$$N_2 \xrightarrow{t > 5/\lambda_2} \frac{\lambda_1}{\lambda_2} N_1^0 e^{-\lambda_1 t} = \frac{\lambda_1}{\lambda_2} N_1$$

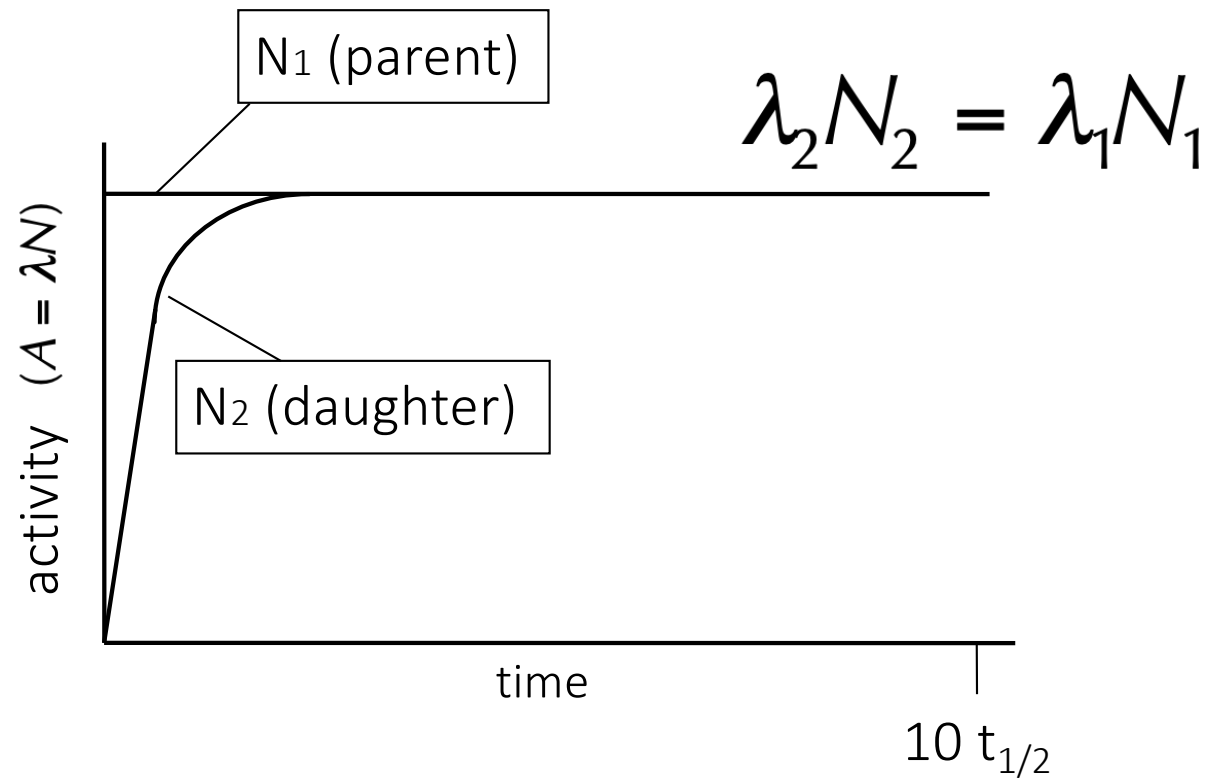
$$\lambda_2 N_2 = \lambda_1 N_1$$

$$A_2 = A_1$$

It takes about 5 mean-lives of N_2 to reach secular equilibrium, in which the ratio of parent to intermediate daughter is fixed. After this point the initial amount of N_2 in the system no longer matters.

Note the N_3 does not participate in secular equilibrium, it just accumulates.

Secular Equilibrium



Secular Equilibrium

$$\lambda_2 N_2 = \lambda_1 N_1$$

- Note that the above equation for secular equilibrium assume no other source and no other removal mechanism. Nevertheless, we know that this is generally not the case in the ocean.
- However, this is what makes radionuclides so useful!
- For example, what if the daughter nuclide is removed by processes other than decay (e.g., scavenging)?

Secular Equilibrium – use of decay series

Element	U-238 Decay Series					Th-232 Decay Series					U-235 Decay Series				
Uranium	U-238 4.47×10^9 y		U-234 2.48×10^5 y	Soluble, conservative							U-235 7.04×10^8 y				
Protactinium		Pa-234 1.18 min										Pa-231 3.25×10^4 y			
Thorium	Th-234 24.1 d		Th-230 7.52×10^4 y	Particle reactive		Th-232 1.40×10^{10} y		Th-228 1.91 y			Th-231 25.5 h		Th-227 18.7 d		
Actinium							Ac-228 6.13 h					Ac-227 21.8 y			
Radium			Ra-226 1.62×10^3 y	Soluble, from sediment		Ra-228 5.75 y		Ra-224 3.66 d					Ra-223 11.4 d		
Francium															
Radon			Rn-222 3.82 d	Inert gas				Rn-220 55.6 s					Rn-219 3.96 s		
Astatine															
Polonium			Po-218 3.05 min		Po-214 1.64×10^{-4} s	Po-210 138 d		Po-216 0.15 s		Po-212 3.0×10^{-7} s			Po-215 1.78×10^{-3} s		
Bismuth				Bi-214 19.7 min		Bi-210 5.01 d			Bi-212 60.6 min					Bi-211 2.15 min	
Lead	Rel. insoluble		Pb-214 26.8 min		Pb-210 22.3 y	Pb-206 stable		Pb-212 10.6 h		Pb-208 stable			Pb-211 36.1 min		Pb-207 stable
Thallium									Tl-208 3.05 min					Tl-207 4.77 min	

Figure 5.18. A chart of the nuclides showing the decay pathways and half lives of isotopes in the three naturally occurring decay chains. Arrows that point downward indicate α decay, in which a nucleus loses two neutrons and two protons, thus decreasing in atomic number by 2 and atomic mass by 4. The arrows that slant upward to the right indicate β decay, in which a neutron in the nucleus becomes a proton (one negative charge is lost from the nucleus), causing an increase in atomic number but little change in atomic mass.

Secular Equilibrium – use of decay series

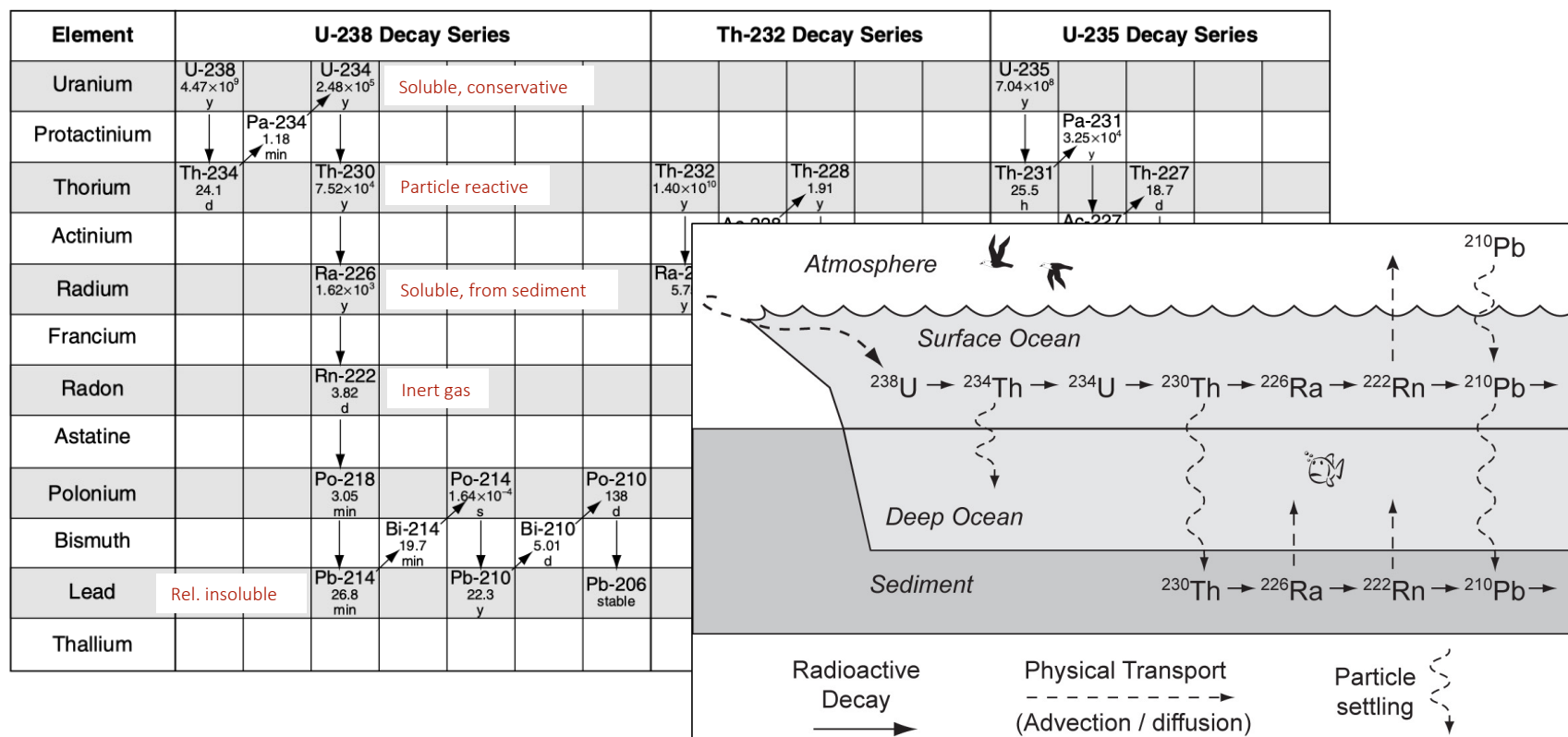
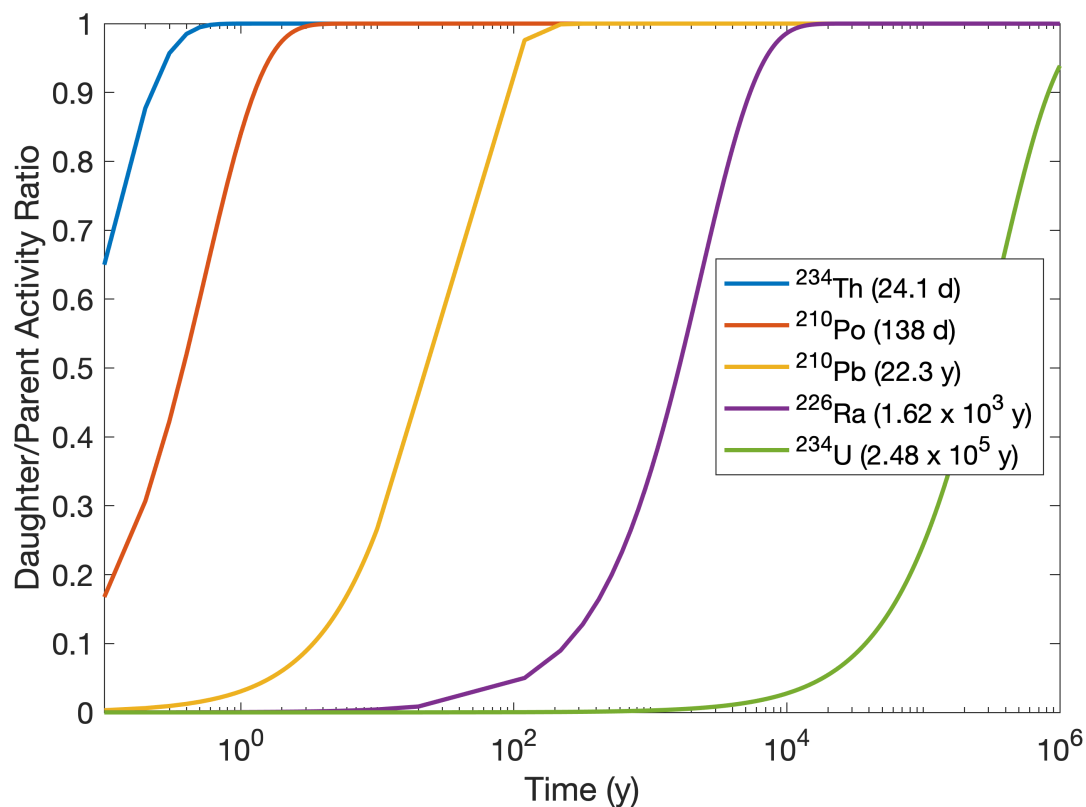


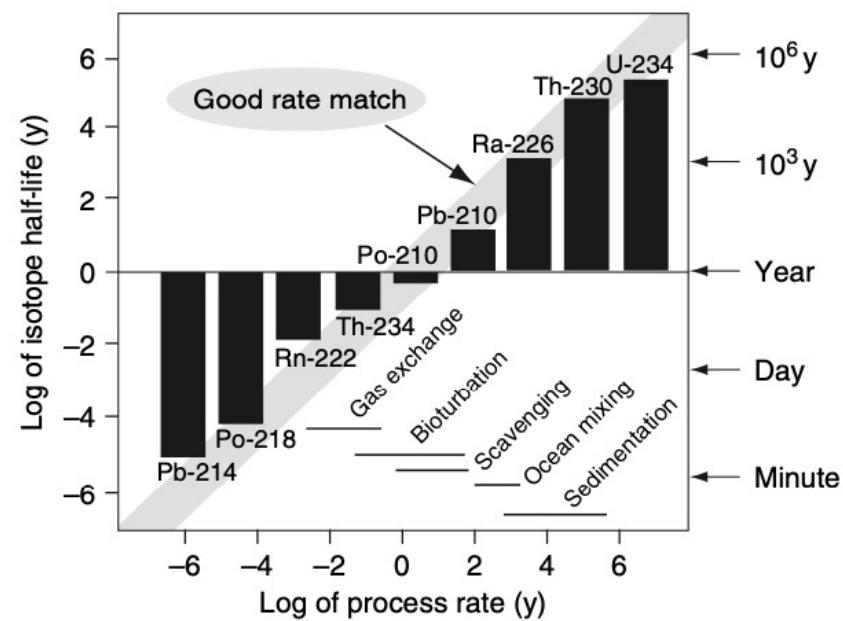
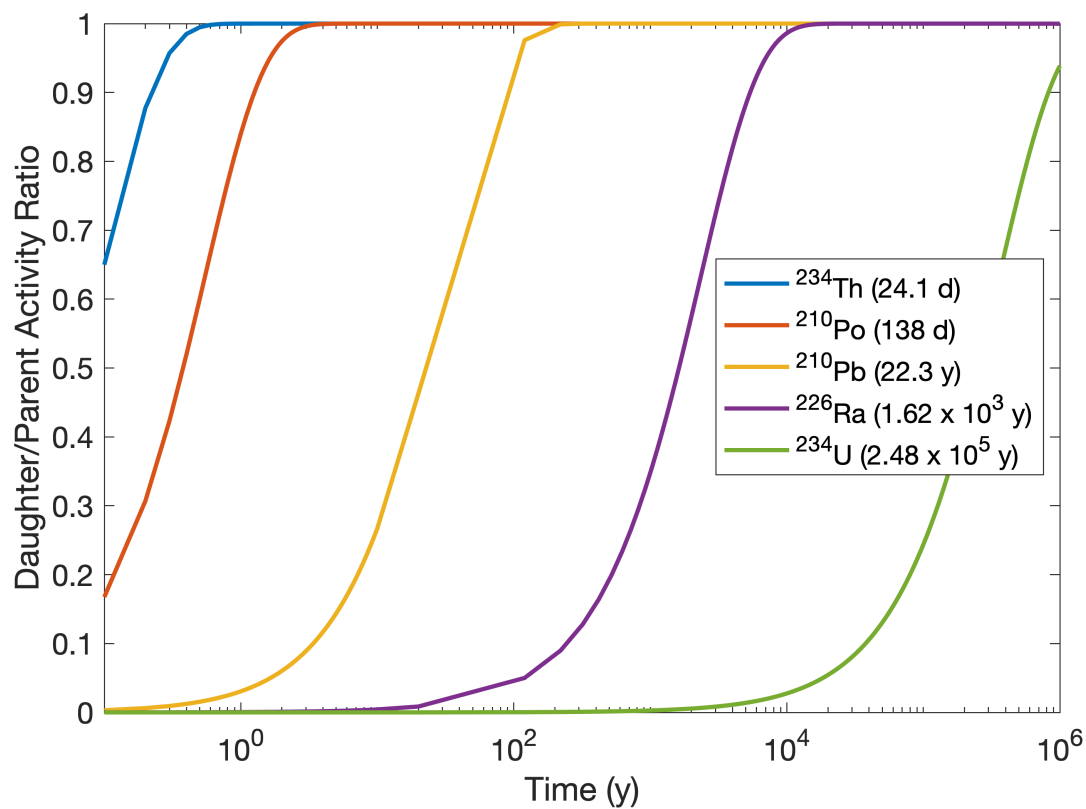
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The longer the half-life of the daughter, the longer it takes to achieve a daughter/parent ratio of 1



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Uranium	U-238 4.47×10^9 y		U-234 2.48×10^5 y			
Protactinium		Pa-234 1.18 min				
Thorium	Th-234 24.1 d		Th-230 7.52×10^4 y			
Actinium						
Radium			Ra-226 1.62×10^3 y			
Francium						
Radon			Rn-222 3.82 d			
Astatine						
Polonium			Po-218 3.05 min	Po-214 1.64×10^{-4} s		Po-210 138 d
Bismuth			Bi-214 19.7 min		Bi-210 5.01 d	
Lead			Pb-214 26.8 min	Pb-210 22.3 y		Pb-206 stable
Thallium						

Process of interest needs to be matched to half-life



Isotopes that have been used in Oceanography

These tracers have a range of origins, chemistries and half lives.

		steady state		transient	
Isotope	t1/2 (y)	Origin		Weapons Testing	Other Anthro.
		Cosmic Rays	U+Th Series		
Water Tracers					
¹⁴ C	5,730	✓		✓	
²²⁶ Ra	1,600		✓		
³² Si	250	✓			
³⁹ Ar	270	✓			
¹³⁷ Cs	30.2			✓	
⁹⁰ Sr	28.6			✓	
³ H	12.33	✓		✓	
⁸⁵ Kr	10.7			✓	✓
²²⁸ Ra	5.8		✓		
⁷ Be	0.15	✓			
²²² Rn	0.01		✓		
Particulate Tracers					
²³⁹ Pu	24,400			✓	✓
²³⁰ Th	75,440		✓		
²⁴⁰ Pu	6,540			✓	✓
²¹⁰ Pb	22.3		✓		
²²⁸ Th	1.9		✓		
²¹⁰ Po	0.38		✓		
²³⁴ Th	0.07		✓		

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^{234}Th and ^{238}U in the Upper Ocean

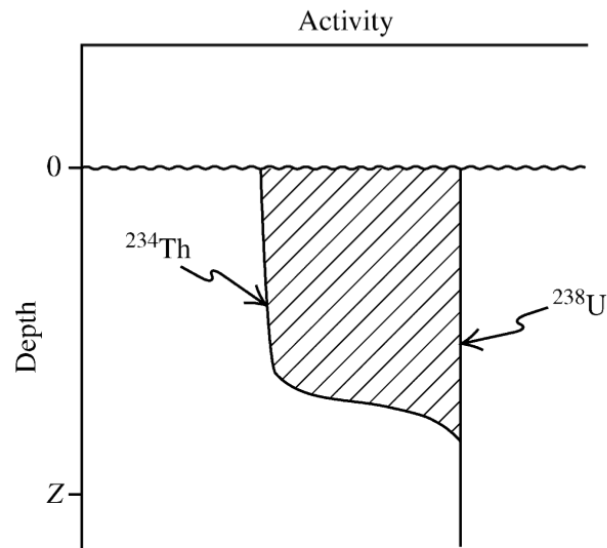


Figure 4 Schematic illustration of concentration profiles of ^{234}Th and ^{238}U in the upper water column. The integrated deficit of ^{234}Th (hatched area) is used to compute the flux of particulate ^{234}Th exported from surface waters.

- ^{238}U and ^{234}Th in secular equilibrium
- Deficit of ^{234}Th ($t_{1/2} = 24.1$ days) relative to its source (i.e., ^{238}U) is due to scavenging
- Th removed from solution, sorbed onto particles

^{234}Th Removal Rate

$$\frac{dN_{^{234}\text{Th}}}{dt} = \underbrace{N_{^{238}\text{U}}\lambda_{^{238}\text{U}}}_{\text{Production}} - \underbrace{N_{^{234}\text{Th}}\lambda_{^{234}\text{Th}}}_{\text{Decay}} - \underbrace{N_{^{234}\text{Th}}\kappa}_{\text{Removal}} \pm \nu$$

Physical processes

$$\frac{dN_{\text{Th}}}{dt} = A_{\text{U}} - A_{\text{Th}} - N_{\text{Th}}\kappa \pm \nu$$

- Assume negligible advection and diffusion
- Multiply by λ_{Th} to convert to activity
- Assume steady state: $dA_{\text{Th}}/dt = 0$

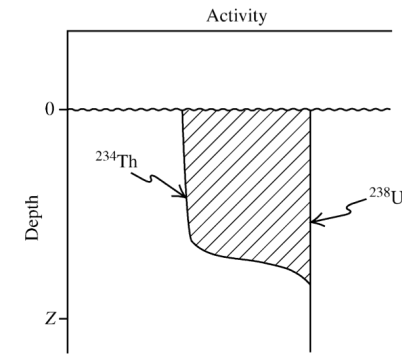
$$\frac{dA_{\text{Th}}}{dt} = 0 = \lambda_{\text{Th}}(A_{\text{U}} - A_{\text{Th}}) - A_{\text{Th}}\kappa$$

$$A_{\text{Th}}\kappa = \lambda_{\text{Th}}(A_{\text{U}} - A_{\text{Th}}) \quad \leftarrow \text{Removal rate (dpm L}^{-1}\text{ d}^{-1}\text{)}$$

^{234}Th Flux Calculation

$$\text{Flux} = \kappa A_{\text{Th}} z = (A_{\text{U}} - A_{\text{Th}}) \lambda_{\text{Th}} z 10^3$$

$\text{dpm m}^{-2} \text{d}^{-1}$ water depth of mixed layer dpm L^{-1} d^{-1} m L m^{-3}



- Deficit of thorium integrated over the water column, gives the flux of thorium.
- To calculate flux of any element or compound in the particles, multiply ^{234}Th flux by the ratio (in the particles) of ^{234}Th to the species of interest.
 - *e.g.*, C flux = ^{234}Th flux * (C / ^{234}Th)_{sinking particles}
- Flux of thorium related to flux of particles, and therefore related to flux of C.